Name			
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Date	Conduction	Record	Viva
Allotted Marks			
Obtained Marks			

EXPERIMENT 3

Develop a Map Reduce program that mines weather data and displays appropriate messages indicating the weather conditions of the day

Aim: To develop a Map Reduce program that mines weather data and displays appropriate messages indicating the weather conditions of the day.

Theory:

Weather data is typically huge (hourly readings from thousands of sensors). MapReduce allows efficient, scalable processing of such data to extract

- Maximum or minimum temperatures
- Rainfall status
- Wind speeds
- Condition messages (e.g., “Hot Day,” “Rainy Day”)

The Mapper will:

- Read each line of weather data.
- Extract the date, temperature, humidity, and rainfall.
- Output the date as the **key** and weather metrics as the **value**.

The Reducer will:

- Receive all weather records grouped by date.
- Compute maximum temperature, average humidity, total rainfall.
- Based on these, generate a descriptive weather message.

Procedure:

Procedure:

1. **Start the sandbox, once all the applications are running , open putty & login**

2. Type

[maria_dev@sandbox-hdp ~]$ vi pro3.py

To enter the code type insert

```
from mrjob.job import MRJob
from mrjob.step import MRStep

class WeatherAnalysis(MRJob):

 def mapper(self, _, line):
  """
  Mapper function to extract weather conditions from the dataset.
 """
 parts = line.strip().split(',')
 if len(parts) != 4:  # Skip malformed lines
return

  date, temperature, humidity, condition = parts

  try:
temperature = float(temperature)
humidity = float(humidity)
   except ValueError:
return  # Skip lines where values are not numbers

   # Assign a weather message based on the condition
   if condition.lower() in ["sunny", "clear"]:
message = "It's a bright and sunny day! Wear sunscreen. ☀"
  elif condition.lower() in ["rainy", "drizzle"]:
message = "Take an umbrella, it's going to rain! ☂ "
   elif condition.lower() == "cold":
message = "It's a cold day! Wear warm clothes. ❄ 🧥"
  elif condition.lower() == "hot":
message = "Stay hydrated! It's really hot today. ☀"
  elif condition.lower() == "stormy":
message = "Be cautious! A storm is expected. ⛈"
   else:
message = "Weather condition unknown. Stay prepared!"

 yield date, message

 def reducer(self, date, messages):
  """
  Reducer function to output final weather messages.
   """
   for message in messages:
     yield date, message
```

if __name__ == '__main__':
WeatherAnalysis.run()

Press esc

:wq

[maria_dev@sandbox-hdp ~]$ cat<<EOF> data.txt

>2025-03-10,30,70,sunny

>2025-03-11,25,85,rainy

>EOF

To check the result type the following

[maria_dev@sandbox-hdp ~]$ python3 pro3.py data.txt

```
[maria_dev@sandbox-hdp ~]$ vi pro3.py
[maria_dev@sandbox-hdp ~]$ cat << EOF > data.txt
> 2025-03-10,30,70,sunny
> 2025-03-11,25,85,rainy
> EOF
[maria_dev@sandbox-hdp ~]$ python3 pro3.py data.txt
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/pro3.maria_dev.20250417.072441.293683
Running step 1 of 1...
job output is in /tmp/pro3.maria_dev.20250417.072441.293683/output
Streaming final output from /tmp/pro3.maria_dev.20250417.072441.293683/output...
"2025-03-10"    "It's a bright and sunny day! Wear sunscreen. \ud83d\ude0e"
"2025-03-11"    "Take an umbrella, it's going to rain! \u2614"
Removing temp directory /tmp/pro3.maria_dev.20250417.072441.293683...
[maria_dev@sandbox-hdp ~]$
```

Results:

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